Lecture - 12

23/10/2025

Adjucate of a Matrix

Let A be an $n \times n$ matrix. Let $C = (c_{ij})$ be the matrix such that

$$c_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Then C^T is called the adjucate of A.

The number c_{ij} is called the ij^{th} cofactor of A.

The matrix C is called the cofactor matrix of A.

Theorem: Let A be an $n \times n$ matrix and c be the cofactor matrix of A. Then $AC^T = C^T A = \det(A)\, I_n$.

Proof Let $B = AC^T$ and denote the ij entry of B by b_{ij}. Then

$$\text{(A)} \quad b_{ij} = \sum_{k=1}^{n} a_{ik} \cdot c_{jk} = \sum_{k=1}^{n} a_{ik} (-1)^{j+k} \det(A_{jk})$$

What we need to show is that $B = \det(A) I_n$. That is, $b_{11} = b_{22} = \cdots = b_{nn} = \det(A)$ and $b_{ij} = 0$ if $i \neq j$.

* When $i = j$, we have

$$b_{ii} = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det(A_{ik}) = \det(A).$$

* When $i \neq j$, we consider the matrix D obtained by replacing j^{th} row of A by i^{th} row of A.

$$A = \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix} \qquad D = \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ \vdots \\ R_i \\ \vdots \\ R_n \end{pmatrix} \begin{array}{l} \\ \\ \\ \\ \leftarrow j^{th} \text{ row} \\ \\ \\ \end{array}$$

Then $\det(D) = 0$. Also,

$$0 = \det(D) = \sum_{l=1}^{n} (-1)^{j+l} d_{jl} \det(D_{jl})$$

$$= \sum_{l=1}^{n} (-1)^{j+l} d_{jl} \det(A_{jl}) = \sum_{l=1}^{n} (-1)^{j+l} a_{il} \det(A_{jl})$$

Thus, from (A), we obtain $b_{ij} = 0$ for $i \neq j$ $\square$

$\boxed{\text{Corollaries}}$

(1) If A is invertible then

$$A^{-1} = C^{T} \cdot \frac{1}{\det(A)}.$$

(2) You can expand the determinant via any column as well (use transpose).

HW Project. Learn about Cramer's rule and write a report on it not exceeding 3 pages.